

BANK LOAN DATA ANALYSIS

Project Overview

The Bank Loan Analytics Dashboard is an interactive Power BI solution developed to analyze loan portfolio performance, borrower characteristics, lending trends, and loan repayment behavior. The dashboard transforms raw loan data into meaningful business insights that support data-driven decision-making for financial institutions.

The primary objective of this project is to evaluate lending performance, monitor portfolio health, identify risk factors, and provide stakeholders with actionable insights through dynamic visualizations and interactive reporting.

Dashboard Structure

The dashboard consists of four interconnected pages:

1. Introduction
2. Executive Summary
3. Overview Analysis
4. Detailed Loan Records

1. Introduction Page

The Introduction page provides a high-level overview of the project and explains the purpose of the dashboard.

Key Objectives

- Analyze overall loan portfolio performance.
- Evaluate borrower repayment behavior.
- Identify good and bad loan segments.
- Monitor lending trends and risk indicators.
- Support strategic lending decisions through data-driven insights.

Business Benefits

- Improved loan monitoring.
- Better credit risk assessment.
- Enhanced portfolio management.
- Increased operational efficiency.
- Faster decision-making through interactive reporting.

2. Executive Summary Page

The Executive Summary page presents the most critical Key Performance Indicators (KPIs) required for evaluating overall lending performance.

Key Metrics

Total Loan Applications

- Total applications received: 38.6K
- Indicates overall demand for loan products.

Total Funded Amount

- Total loan amount disbursed: \$435.8M
- Represents the total lending volume.

Total Amount Received

- Total repayments collected: \$473.1M
- Measures cash inflow and portfolio recovery performance.

Average Interest Rate

- Average lending interest rate: 12.05%
- Helps evaluate pricing strategy and lending profitability.

Average Debt-to-Income Ratio (DTI)

- Average DTI: 13.33%
- Indicates borrower repayment capacity and financial stability.

Loan Quality Analysis

Good Loans

- Represent approximately 86.2% of total applications.
- Include Fully Paid and Current loans.
- Contribute significantly to portfolio stability and revenue generation.

Bad Loans

- Represent approximately 13.8% of total applications.
- Include Charged-Off loans.
- Highlight areas requiring risk mitigation and collection efforts.

Loan Status Analysis

The loan status matrix provides a detailed breakdown of:

- Total applications

- Funded amount
- Amount received
- MTD performance
- Average interest rate
- Average DTI

This analysis helps assess the health and performance of each loan category.

3. Overview Analysis Page

The Overview page focuses on identifying lending patterns, borrower demographics, and loan distribution trends.

Monthly Loan Application Trend

A line chart visualizes loan applications across different months.

Insights:

- Identifies seasonal lending patterns.
- Tracks application growth over time.
- Helps forecast future loan demand.

State-wise Loan Distribution

A geographical map displays loan applications by state.

Insights:

- Highlights high-performing regions.
- Identifies geographical concentration of lending activities.
- Supports regional business strategy.

Loan Term Analysis

A donut chart compares loan distribution across different loan terms.

Insights:

- Reveals borrower preferences regarding repayment duration.
- Assists in product planning and loan structuring.

Employment Length Analysis

A bar chart analyzes loan applications by employment tenure.

Insights:

- Longer employment history is associated with higher loan application volumes.
- Helps assess borrower stability.

Loan Purpose Analysis

A bar chart categorizes loans based on borrower purpose.

Insights:

- Debt consolidation and credit card-related loans represent major borrowing purposes.
- Supports product and marketing decisions.

Home Ownership Analysis

A treemap visualizes loan distribution across home ownership categories.

Insights:

- Compares borrowing behavior among homeowners, renters, and mortgage holders.
- Helps understand borrower profiles.

4. Detailed Loan Records Page

The Details page provides transaction-level loan information for in-depth analysis.

Available Fields

- Loan ID
- Home Ownership
- Grade
- Purpose
- Sub Grade
- Funded Amount
- Received Amount
- Interest Rate
- Installment
- Issue Date

Interactive Filters

Users can filter data by:

- State
- Grade
- Good vs Bad Loan

This page enables detailed investigation of individual loan records and supports operational decision-making.

Key Business Insights

1. The loan portfolio demonstrates strong performance with over 86% classified as Good Loans.
2. Total repayments exceed the total funded amount, indicating healthy recovery performance.
3. Debt consolidation is the primary reason for loan applications, representing a major customer financing need.
4. Borrowers with longer employment histories account for a significant portion of loan applications.
5. Lending activity is concentrated in specific states, highlighting regional opportunities and risks.
6. The average DTI ratio remains within a manageable range, indicating relatively healthy borrower financial profiles.

Conclusion

The Bank Loan Analytics Dashboard successfully transforms complex loan data into actionable business intelligence. Through comprehensive KPI monitoring, loan quality assessment, borrower analysis, and detailed transaction reporting, the dashboard enables stakeholders to evaluate portfolio performance, identify risks, optimize lending strategies, and make informed business decisions.
